Unit - 2: Data Communication Principles

Signal: Signal is a function that conveys information about the behavior or attributes of some phenomenon. The electromagnetic signals are used to transmit data in data communication. The signal is a function of time, but it can be expressed as function of frequency as well.

Analog and digital signal:

An **analog signal** is one in which the signal intensity varies in a smooth fashion over time. In other words, there are no breaks or discontinuities in the signal. A **digital signal** is one in which the signal intensity maintains a constant level for some period of time and then abruptly changes to another constant level.

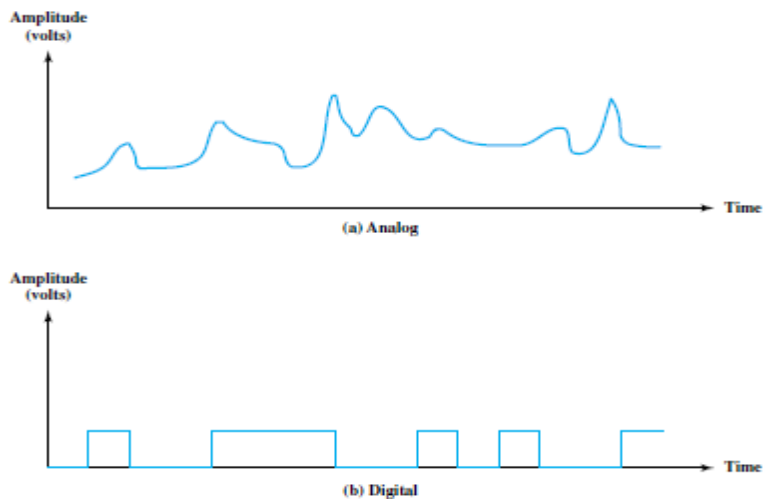
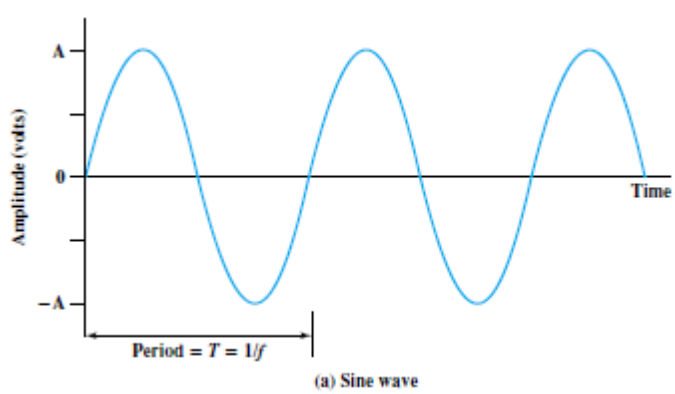


Fig1: Analog and Digital Signal

Periodic and aperiodic signal:

The simplest sort of signal is a **periodic signal**, in which the same signal pattern repeats over time. Otherwise, a signal is **aperiodic**. The sine wave is the fundamental periodic signal. A general sine wave can be represented by three parameters: peak amplitude (*A*), frequency (*f*), and phase (Φ).

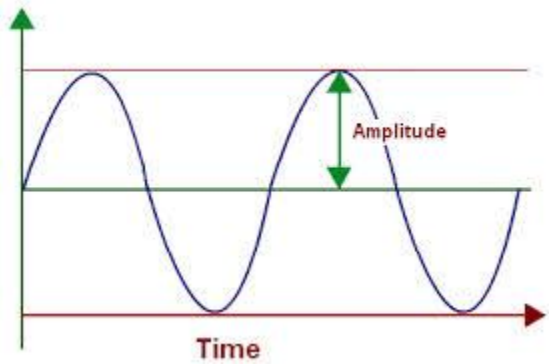


Frequency: Frequency is the number of complete cycles per second. The standard unit of frequency is hertz (Hz).

Period: Time taken by the signal to complete one revolution is known as period:

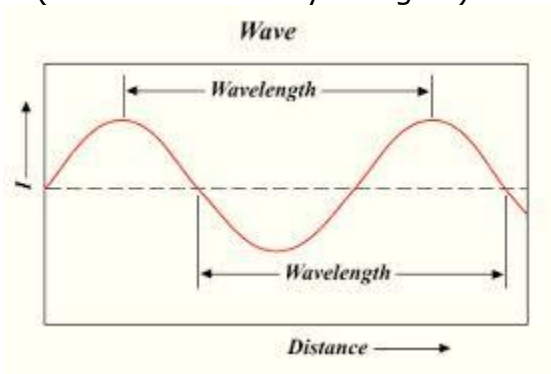
$$\text{Period} = T = 1/f$$

Amplitude: The amplitude is the the power of the signal. The greater the amplitude, the greater the energy carried.



Wavelength: The wavelength of a sinusoidal wave is the spatial period of the wave i.e. the distance over which the wave's shape repeats.

$\lambda = v/f$ (where v is velocity of signal)



Bandwidth:

1) In computer Network Bandwidth is the amount of data that can be carried from one point to another in a given time period (usually a second). This kind of bandwidth is usually expressed in bits (of data) per second (bps).

2) In electronic communication, bandwidth is the width of the range of frequencies that an electronic signal uses on a given transmission medium. Bandwidth is expressed in terms of the difference between the highest-frequency signal component and the lowest-frequency signal component. A typical voice signal has a bandwidth of approximately three kilohertz (3 kHz); an analog television (TV) broadcast video signal has a bandwidth of six megahertz (6 MHz).

Example: An electronic signal uses 20 kHz as its lowest signal component and 40 kHz as highest signal component. Calculate its bandwidth.

Bandwidth = Highest Signal Component – Lowest Signal Component
= 40 kHz – 20 kHz = 20 kHz

Analog and Digital Data:

Analog data refers to information that is continuous; For example, an analog clock that has hour, minute, and second hands gives information in a continuous form; the movements of the hands are continuous. Analog data, such as the sounds made by a human voice, take on continuous values. When someone speaks, an analog wave is created in the air. This can be captured by a microphone and converted to an analog signal or sampled and converted to a digital signal.

Digital data refers to information that has discrete states. Digital data take on discrete values i.e. values changes after some interval. A digital clock that reports the hours and the minutes will change suddenly from 8:05 to 8:06. Data are stored in computer memory in the form of Os and 1s. They can be converted to a digital signal or modulated into an analog signal for transmission across a medium.

Analog Transmission and Digital Transmission:

Both analog and digital signals may be transmitted on suitable transmission media. The way these signals are treated is a function of the transmission system.

Analog transmission is a means of transmitting analog signals without regard to their content; the signals may represent analog data (e.g., voice) or digital data. The analog signal will become weaker (attenuate) after a certain distance. To achieve longer distances, the analog transmission system includes amplifiers that boost the energy in the signal. Unfortunately, the amplifier also boosts the noise components. With amplifiers cascaded to achieve long distances, the signal becomes more and more distorted.

Digital transmission assumes the signal is carrying binary data. A digital signal can be transmitted only a limited distance before attenuation, noise, and other impairment corrupt the data. To achieve greater distances, repeaters are used. A repeater receives the digital signal, recovers the pattern of 1s and 0s, and retransmits a new signal. Thus the attenuation is overcome. The same technique may be used with an analog signal if it is assumed that the signal carries digital data. The repeater recovers the digital data from the analog signal and generates a new, clean analog signal.

	Analog Transmission	Digital Transmission
Analog Signal	Is propagated through amplifiers; same treatment whether signal is used to represent analog data or digital data.	Assumes that the analog signal represents digital data. Signal is propagated through repeaters; at each repeater, digital data are recovered from inbound signal and used to generate a new analog outbound signal.
Digital Signal	Not used	Digital signal represents a stream of 1s and 0s, which may represent digital data or may be an encoding of analog data. Signal is propagated through repeaters; at each repeater, stream of 1s and 0s is recovered from inbound signal and used to generate a new digital outbound signal.

Table: Characteristics of analog and digital transmission

Synchronous and Asynchronous Transmission:

Synchronous transmission: Synchronous transmission sends data as one long bit stream or block of data. There are no gaps in transmission; each bit is sent one after the other. The receiver counts the bits and reconstructs bytes. Timing is maintained through clock as there are no start and stop bits and no gaps. Accuracy is dependent on the receiver keeping an accurate count of the bits as they come in. Synchronous transmission is faster than asynchronous because fewer bits have to be transmitted; i.e. only data bits and no extra control bits. For this reason it is the choice for network communications links.

Asynchronous transmission: In Asynchronous transmission signal timing is not required; signals are sent in an agreed pattern of bits and if both ends are agreed on the pattern then communication can take place.
Bits are grouped together and consist of both data and control bits. Here, the data is preceded by a start bit, usually binary 0, the data (byte) is then sent and a stop bit or bits are added to the end. Each byte to be sent now incorporates extra control data. In addition to the control data small gaps are inserted between each chunk to distinguish each group.

Asynchronous transmission is relatively slow due to the increased number of bits and gaps. It is a cheap and effective form of serial transmission and is particularly suited for low speed connections such as keyboard and mouse.

Transmission Impairments

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. For analog signals, these impairments can degrade the signal quality. For digital signals, bit errors may be introduced, such that a binary 1 is transformed into a binary 0 or vice versa.

The most significant impairments are:

i. Attenuation

Attenuation means a loss of energy. When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium. That is why a wire carrying electric signals gets warm, if not hot, after a while. Some of the electrical energy in the signal is converted to heat. To compensate for this loss, amplifiers are used to amplify the signal.

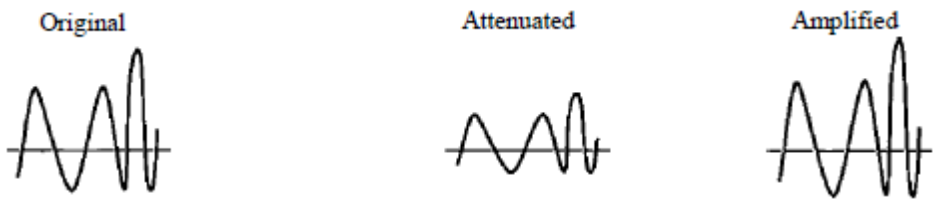


Fig: Attenuation and amplification

Attenuation is measured in terms of decibel. The attenuation of signal transmitted from point 1 to 2 is given by following relation

$$dB = 10\log_{10}P2/P1$$

Here P1 and P2 are power of signal at point 1 and 2 respectively.

Problem 1

Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that *P2 = P1/2.* In this case, the attenuation (loss of power) can be calculated as

$$10\log_{10}P2/P1 = 10\log_{10}0.5P1/P1 = 10\log_{10}0.5 = 10(-0.3) = -3dB$$

A loss of 3 dB (-3 dB) is equivalent to losing one-half the power.

Problem 2

A signal travels through an amplifier, and its power is increased 10 times. This means that P2= 1OP1. In this case, the amplification (gain of power) can be calculated as

$$10\log_{10}\frac{P_2}{P_1} = 10\log_{10}\frac{10P_1}{P_1} = 10\log_{10}10 = 10(1) = 10\text{ dB}$$

ii. Distortion

Distortion means that the signal changes its form or shape. Distortion can occur in a composite signal made of different frequencies. Each signal component has its own propagation speed through a medium and, therefore, its own delay in arriving at the final destination. Differences in delay may create a difference in phase if the delay is not exactly the same as the period duration. In other words, signal components at the receiver have phases different from what they had at the sender. The shape of the composite signal is therefore not the same

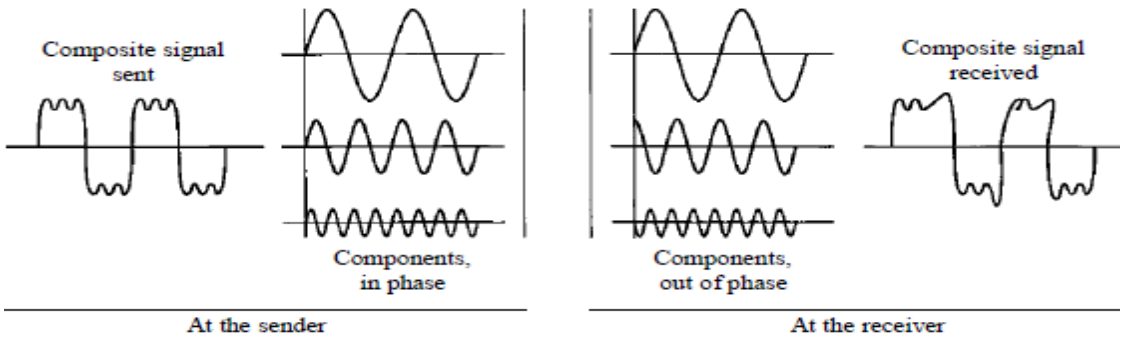


Fig: Distortion

iii. Noise

Noise is another cause of impairment. Several types of noise, such as **thermal noise, induced noise, crosstalk, and impulse noise**, may corrupt the signal. **Thermal noise** is the random motion of electrons in a wire which creates an extra signal not originally sent by the transmitter. **Induced noise** comes from sources such as motors and appliances. These devices act as a sending antenna, and the transmission medium acts as the receiving antenna. **Crosstalk** is the effect of one wire on the other. One wire acts as a sending antenna and the other as the receiving antenna. **Impulse** noise is a spike (a signal with high energy in a very short time) that comes from power lines, lightning, and so on.

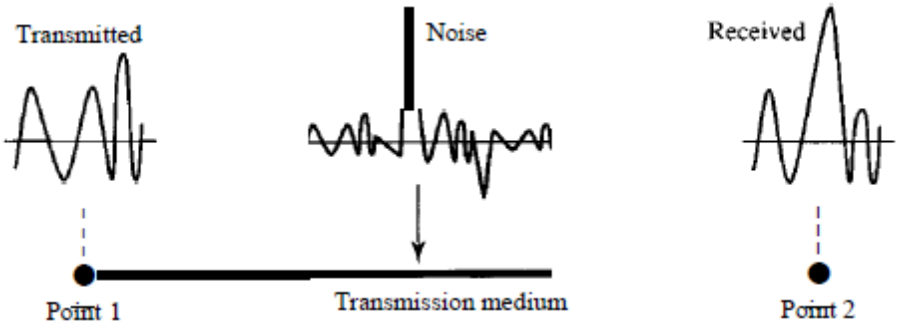


Fig: Noise and Signal

Signal to Noise Ratio:

Signal to noise ratio is defined as ratio of average power of signal to average power of noise.

SNR= Average power of signal/ Average power of noise

Because SNR is the ratio of two powers, it is often described in decibel units, SNR_{dB}, defined as

$$SNR_{dB} = 10\log_{10}SNR$$

Example: The power of a signal is 10 mW and the power of the noise is 1 μW; what are the values of SNR and SNR_{dB}?

The values of SNR and SNR_{dB} can be calculated as follows:

SNR = 10,000 μW/1 μW = 10,000

SNRdB = 10 $\log_{10}$10,000 = 10 $\log_{10}10^4$ = 10 X 4 = 40dB

Calculation of Thermal Noise

The amount of thermal noise to be found in a bandwidth of 1 Hz in any device or conductor is

$$N_0 = kT(\text{W/Hz})$$

Where,
N_0 = noise power density in watts per 1 Hz of bandwidth
T = temperature, in Kelvin (1·C = 273K)
k = Boltzmann's constant = 1.38 X 10^{-23} J/K

Example: Room temperature is 17·C .At this temperature, the thermal noise power density is
N_0 = 1.38 X 10^{-23} X 290 = 4 X 10^{-21} W/Hz

Channel Capacity

In data communication **channel capacity** is the highest possible rate of information that can be reliably transmitted over a communications channel. **Data rate/channel capacity** depends on three factors:

1. The bandwidth available
2. The level of the signals we use
3. The quality of the channel (the level of noise)

Two theoretical formulas were developed to calculate the data rate: one by Nyquist for a noiseless channel, another by Shannon for a noisy channel.

A) Nyquist Bandwidth:

For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate

$$\text{BitRate} = 2B\log_2 M$$

B is the bandwidth of the channel
M is the number of signal levels used to represent data i.e. M= 2^n where n is no of bits per character.

Examples

Example 1: Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels (1 bit per level). The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

Example 2: Consider the same noiseless channel transmitting a signal with four signal levels (for each level/character, we send 2 bits). The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \text{ X } \log_2 4 = 12{,}000 \text{ bps}$$

Example 3: We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

Solution:

We can use the Nyquist formula as shown:

$$\text{BitRate} = 2B\log_2 M$$
$$265{,}000 = 2 \times 20{,}000 \times \log_2 M$$
$$\log_2 M = 6.625$$
$$M = 2^{6.625} = 98.7 \text{ levels}$$

Since this result is not a power of 2, we need to either increase the number of levels or reduce the bit rate. If we have 128 levels, the bit rate is 280 kbps. If we have 64 levels, the bit rate is 240 kbps.

B) Shannon Channel Capacity:

In reality, we cannot have a noiseless channel; the channel is always noisy. In 1944, Claude Shannon introduced a formula, called the Shannon capacity, to determine the theoretical highest data rate for a noisy channel:

$$\text{Capacity} = B\log_2 (1 + \text{SNR})$$

B is the bandwidth of the channel,
SNR is the signal-to noise ratio

Example1: A telephone line normally has a bandwidth of 3000 Hz assigned for data communications. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as

$$C = B \log_2 (1 + \text{SNR}) = 3000 \log2 (l + 3162) = 3000 \log_2 3163$$
$$= 3000 \times 11.62 = 34{,}860 \text{ bps}$$

Example2: The signal-to-noise ratio is often given in decibels. Assume that SNR_{dB} is 36 and the channel bandwidth is 2 MHz. The theoretical channel capacity can be calculated as

$$SNR_{dB} = 10\log_{10}SNR$$
$$\log_{10}SNR = SNR_{dB}/10$$
$$SNR = 10^{SNRdB/10}$$
$$SNR = 10^{3.6} = 3981$$

Now,

$$C = B \log_2 (1 + \text{SNR}) = 2 \times 10^6 \times \log_2 3982 = 24 \text{ Mbps}$$

Data Encoding

Data Encoding is the process of representing the information into signals. Both analog and digital information can be encoded as either analog or digital signals. The particular encoding that is chosen depends on the specific requirements to be met and the media and communications facilities available.

I. Digital Data and Digital Signals

The nature of the actual signals representing the bits on the media will depend on the signaling method in use. Some methods may use one attribute of signal to represent a single 0 and use another attribute of signal to represent a single 1.

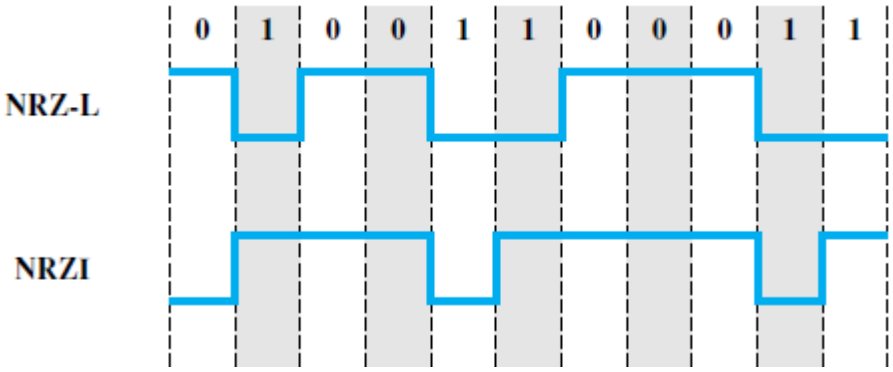


Fig: NRZ-L and NRZ-I Encoding

Non-Return to Zero-Level (NRZ-L): In this encoding scheme, a 0 may be represented by one voltage level during the bit time and a 1 might be represented by a different voltag during the bit time. In the above figure 0 is represented by high voltage level and 1 is represented by low voltage level. It can also be represented in opposite order i.e 0(low) and 1(high).

Non-Return to Zero-Inverted (NRZ-I): In this encoding scheme, 0 is represented by continuity of signal level of prevoius bit time and 1 is represented by transition of state each time 1 occurs in the bitstream.

Manchester Encoding: In Manchester Encoding a 0 is indicated by a high to low voltage transition in the middle of the bit time. For a 1 there is a low to high voltage transition in the middle of the bit time as shown in figure.

Differential Manchester:

Always a transition in middle of interval
0 = transition at beginning of interval
1 = no transition at beginning of interval

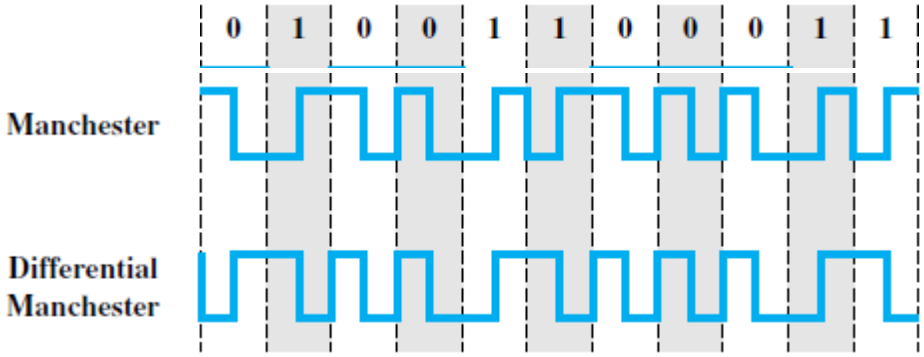


Fig: Manchester and differential Manchester encoding

Digital Data and Analog Signals

The encoding of analog data into analog signals is knowns as modulation. Some of the modulation techniques are as follows.

i. Amplitude Shift Keying (ASK)

In ASK, the two binary values are represented by two different amplitudes of the carrier frequency. Commonly, one of the amplitudes is zero; that is, one binary digit is represented by the presence of constant amplitude of the carrier wave and the other by the absence of the carrier. The resulting transmitted signal for one bit time is:

$$\textbf{ASK} \qquad s(t) = \begin{cases} A\cos(2\pi f_c t) & \text{binary } 1 \\ 0 & \text{binary } 0 \end{cases}$$

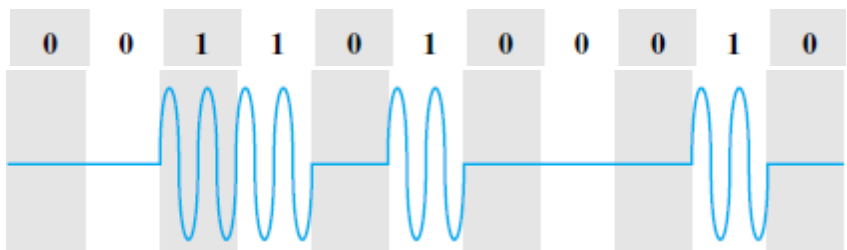


Fig: ASK

ii. Frequency Shift Keying (FSK)

The most common form of FSK is binary FSK (BFSK), in which the two binary values are represented by two different frequencies near the carrier frequency. The resulting transmitted signal for one bit time is:

$$\textbf{BFSK} \qquad s(t) = \begin{cases} A\cos(2\pi f_1 t) & \text{binary } 1 \\ A\cos(2\pi f_2 t) & \text{binary } 0 \end{cases}$$

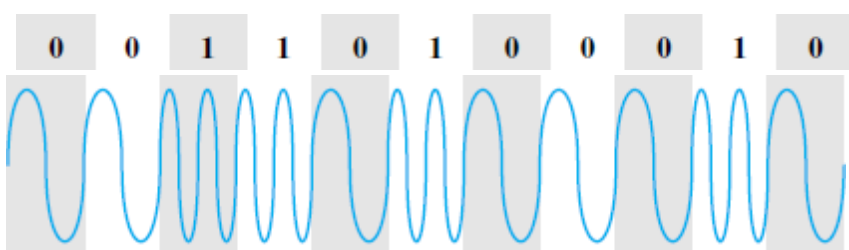


Fig: BFSK

iii. Phase Shift Keying (PSK)

The simplest scheme uses two phases to represent the two binary digits and is known as binary phase shift keying. The resulting transmitted signal for one bit time is

$$\textbf{BPSK} \qquad s(t) = \begin{cases} A\cos(2\pi f_c t) \\ A\cos(2\pi f_c t + \pi) \end{cases} = \begin{cases} A\cos(2\pi f_c t) & \text{binary } 1 \\ -A\cos(2\pi f_c t) & \text{binary } 0 \end{cases}$$

A phase shift of 180° is equivalent to flipping the sine wave or multiplying it by -1.

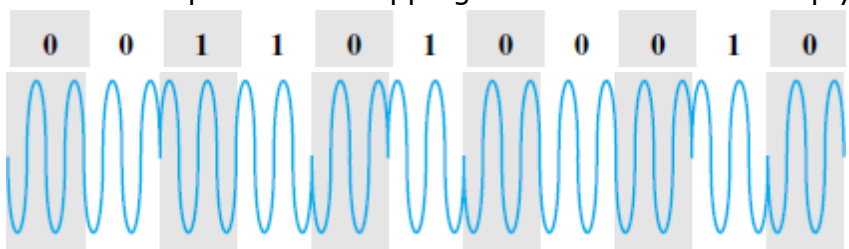
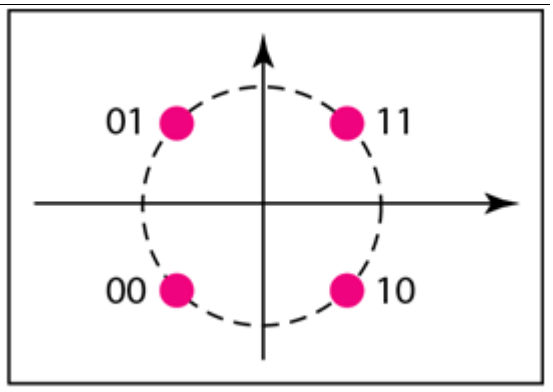


Fig: BPSK

iv. Quadrature Phase Shift Keying (QPSK)

More efficient use of bandwidth can be achieved if each signaling element represents more than one bit. For example, instead of a phase shift of 180°, as allowed in BPSK, a common encoding technique, known as Quadrature phase shift keying (QPSK), uses phase shifts separated by multiples of $\frac{\pi}{2}$ (90°). Thus each signal element represents two bits rather than one.

$$\textbf{QPSK} \qquad s(t) = \begin{cases} A\cos\left(2\pi f_c t + \frac{\pi}{4}\right) & 11 \\ A\cos\left(2\pi f_c t + \frac{3\pi}{4}\right) & 01 \\ A\cos\left(2\pi f_c t - \frac{3\pi}{4}\right) & 00 \\ A\cos\left(2\pi f_c t - \frac{\pi}{4}\right) & 10 \end{cases}$$



Multiplexing:

Multiplexing is a technique by which different analog and digital streams of transmission can be simultaneously processed over a shared link. Multiplexing divides the high capacity medium into low capacity logical medium which is then shared by different streams. When more than one sender tries to send over single medium, a device called Multiplexer divides the physical channel and allocates one to each. On the other end of communication, a De-multiplexer receives data from a single medium and identifies each and sends to different receivers.

Frequency Division Multiplexing

FDM is an analog technology. FDM divides the spectrum or carrier bandwidth in logical channels and allocates one user to each channel. Each user can use the channel frequency independently and has exclusive access of it. All channels are divided such a way that they do not overlap with each other. Channels are separated by guard bands. Guard band is a frequency which is not used by either channel.

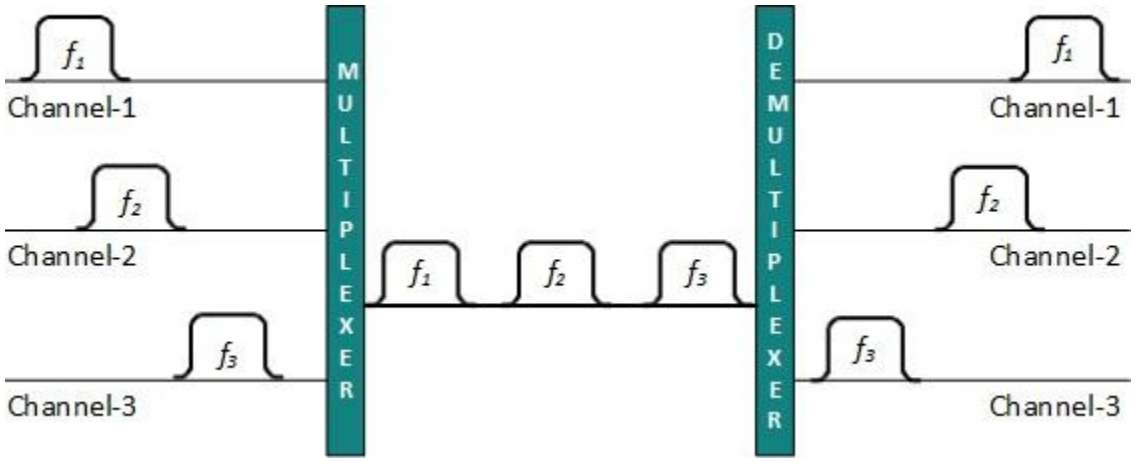


Fig: *Frequency Division Multiplexing*

Time Division Multiplexing

TDM is applied primarily on digital signals but can be applied on analog signals as well. In TDM the shared channel is divided among its user by means of time slot. Each user can transmit data within the provided time slot only. Digital signals are divided in frames, equivalent to time slot i.e. frame of an optimal size which can be transmitted in given time slot. TDM works in synchronized mode.

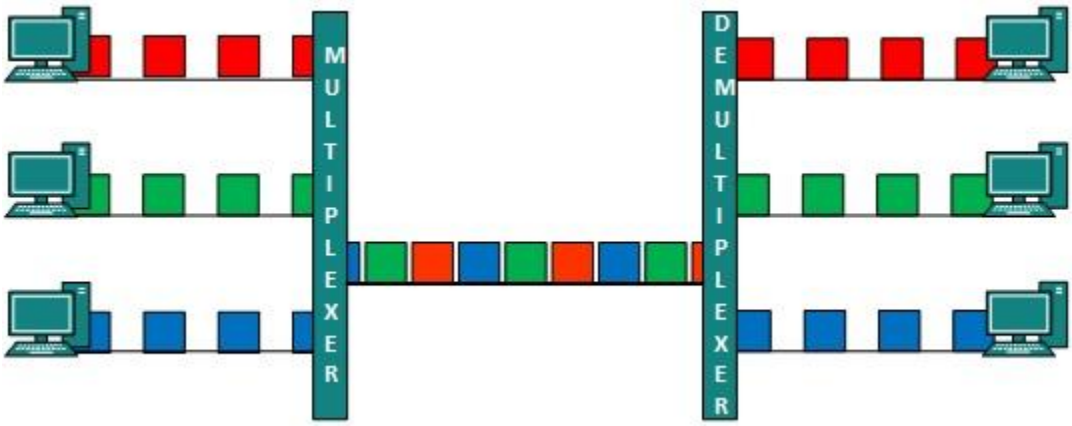


Fig: Time Division Multiplexing

Wavelength Division Multiplexing

Light has different wavelength (colors). In fiber optic mode, multiple optical carrier signals are multiplexed into on optical fiber by using different wavelengths. This is an analog multiplexing technique and is done conceptually in the same manner as FDM but uses light as signals.

Fig*: Wavelength Division Multiplexing*

Further, on each wavelength Time division multiplexing can be incorporated to accommodate more data signals.